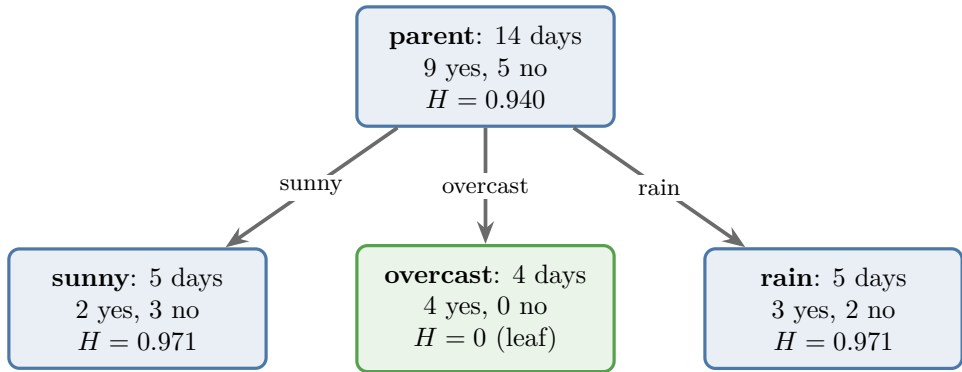




$$\text{weighted entropy of children} = \frac{5}{14}(0.971) + \frac{4}{14}(0) + \frac{5}{14}(0.971) = 0.694$$

$$\text{information gain} = 0.940 - 0.694 = \mathbf{0.247}$$